

Juan David Muñoz Bolaños

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+43 676 9115145

jmunozbolanos@gmail.com

April 22, 2026

Carl Zeiss Vision International GmbH

Turnstraße 27

73430 Aalen

Germany

Application: Pioneer for Extended Reality (XR) | Ref. JR_1048257-1

as an Optical System Engineer with a deep passion for optical innovation, I see the development of smart glasses as the perfect convergence of high precision optics and holography display technology, a frontier that is the future and I want to be part of it.

Throughout my career I have specialized in bridging optical prototypes to work in technological applications such as holographic sensors. During my PhD at the Medical University of Innsbruck I built and integrated a full holographic and adaptive optics platform spanning Texas Instruments spatial light modulators, laser sources, and FPGA based real time control. This work focused on holographic phase retrieval and wavefront sensing—technologies that are fundamental to characterizing holographic optical elements (HOE) and integrating them into optical microscope.

My background in optics, holography, spatial light modulation, and interferometry offers a direct bridge to the eye box optimization and holographic projection or waveguide light challenges to tooz smart glasses. I drived projects end to end from idea, component selection through commissioning, and I enjoy the kind of interdisciplinary, fast moving environment that pioneering a new product category demands.

The opportunity to shape the future of mobile computing platforms at a world leading optical company like ZEISS is a compelling one. I would welcome the chance to visit the tooz Extended Reality vision experience.

Kind regards,

Juan David Muñoz Bolaños

PhD Candidate, Medical University of Innsbruck

Juan David Muñoz Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

jmunozbolanos@gmail.com | LinkedIn Profile



Summary

Optical System Engineer specializing in holographic display metrology and wavefront correction optical systems. Expertise in holographic modulation, and bridging optical prototypes to implementation and applications.

Professional Experience

Jan'22 | Present **Med. Univ. Innsbruck**

Innsbruck, Austria

Optical System Engineer | PhD Researcher

Adaptive Optical Optimization: Developed a Python driven closed loop control system using spatial light modulators to restore signal intensity; applied to active aberration correction in holographic optical elements.

Holographic Wavefront Metrology: Architected high speed holographic phase retrieval and wavefront sensing routines for optical aberration characterization using a C++, Python, and LabVIEW FPGA instrumentation stack.

Holographic Engine Integration: Synchronized high speed display modules, optofluidic modulators, and CMOS sensors to validate holographic concepts; migrated lab setups into robust, portable holographic display prototypes.

Technical Competencies

Optical System Design & Wavefront Control: Knowledge of waveoptics by python simulation and ray optics by ZEMAX OpticStudio, Tolerance Analysis, Holography, Spatial Light Modulators, and interferometric wavefront sensing.

Industrial System Deployment: C++, Python, LabVIEW FPGA for real time instrument synchronization; experience in Gage R&R validation, production floor commissioning, and V Model development cycles.

Machine Vision Software: Halcon (MVTec), Cognex VisionPro, OpenCV; expertise in pattern matching, OCR, and adaptive segmentation for automated inspection and robotic guidance.

Education

Jan'22 | Jun'26 **Medical University of Innsbruck**

Innsbruck, Austria

Doctor of Philosophy in Adaptive Optics & Precision Metrology

2019 | 2021 **Friedrich Schiller Univ. Jena**

Jena, Germany

Master of Science in Photonics

2012 | 2018 **University of Cauca**

Popayán, Colombia

Bachelor of Science in Physics Engineering

Publications & Awards

Awards: Selected for ZEISS SMT and ASLM summer schools, and SPIE Photonics West (San Francisco) Conference. COLFUTURO 2016 to 2018 (best Colombian professionals).

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills & Hobbies

Ownership & Initiative: Lead project focus from ideas to proof of application, holographic wavefront sensor. **Teamwork & Analysis:** Proven collaboration across optics, hardware, and software disciplines with structured root-cause analysis of defects. **Hobbies:** Enjoying the mountains (Biking, Hiking, learning Skiing) • Bachata Dancing • Harmonica • Vision Systems

Languages

English (Fluent) • German (Intermediate B2) • Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

Prof. Dr. Alexander Jesacher
Medical University of Innsbruck
alexander.jesacher@i-med.ac.at

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

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DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

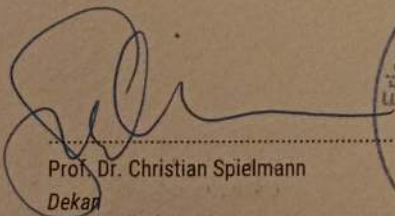
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

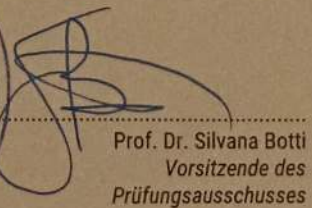
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passports № / Passport No

MUNOZ BOLANOS

COLOMBIAN

Fecha de nacimiento / Date of birth
04 JUN/JUN 1994

Sexo / Sex

Lugar de nacimiento / Place of birth

LA CRUZ COL M

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

Antinostad / Antibodies

G. ANTIOQUIA

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Dear Family

[illegible]

Stadtmagistrat Innsbruck
MA II - Bezirks- und Gemeindeverwaltung
Standesamt und Personenstandsangelegenheiten
Aufenthaltsangelegenheiten
post.aufenthaltsangelegenheiten@innsbruck.gv.at

Zahl: Maglbk/58566/AUFG-NAG/3/1 – 00053337-04

Antragsprotokoll

Herr: **Muñoz Bolaños Juan David**
wohnhaft in: **6020 Pradl, Reichenauer Straße 42/Top 5**
geboren am: **04.06.1994**
Reisedokument-Nr.: **AV410884**
Nationalität: **Kolumbien**
hat am: **13.03.2026**

beim Stadtmagistrat Innsbruck, Referat Aufenthaltsangelegenheiten, einen Zweckänderungsantrag auf Rot-Weiß-Rot - Karte plus eingebracht.

Ein rechtzeitig eingebrachter Verlängerungsauftrag räumt während der gesamten Dauer des Verlängerungsverfahrens dieselbe Rechtsposition ein, die nach Erhalt des letzten Aufenthaltstitels innegehabt wurde. Dies bedeutet auch, dass bei befristet erteilten Aufenthaltstiteln mit unbeschränkten Zugang zum Arbeitsmarkt („Rot-Weiß-Rot Karte plus“ und „Familienangehöriger“) weiterhin während der gesamten Dauer des Verlängerungsverfahrens ein Zugang zum Arbeitsmarkt besteht. Bei darüber hinaus gehenden Fragen einer allfälligen weiterhin bestehenden Beschäftigung wenden Sie sich bitte an das Arbeitsmarktservice (AMS).

Nachzureichende Unterlagen: ***

Die Erledigung des vollständig eingebrachten Antrages wird voraussichtlich innerhalb von Wochen erfolgen.

Persönliche Abholung (erforderlich !) jedenfalls erst ab: 17.05.2026, Zimmer: .Nummer ziehen

Zur Abholung (Abholzeiten: Montag bis Freitag, 08.00 bis 11.30 Uhr) sind mitzubringen:

- **der Reisepass**
- **Aufenthaltstitel in Kartenform**

Für den Bürgermeister:

(Huter)

ausstellende Behörde:

Stadtmagistrat Innsbruck
Abteilung II
Aufenthaltsangelegenheiten

übernommen am: 13.03.2026

Unterschrift:

Bei überprüfenden Amtshandlungen drei Monate nach Ausstellung dieser Bestätigung wird um Rückfrage bei der zuständigen Aufenthaltsbehörde ersucht. (Tel.: 0512/5360-1030/1032/1034/1036)